

Aptimizer: Complete Feature List (V1 to X+)

A consolidated feature index across all eight versions. 424 features total, of which 417 are built and 1 is partially built.

Version	Focus	Features	Built	Status
Aptimizer V1	Core Civil Engineering Planning Platform	126	126	Complete
Aptimizer V2	GIS & Site Intelligence	27	25	In progress
Aptimizer V2.5	Precision, Programme & Assurance	116	116	Complete
Aptimizer V3	AI Engineering Platform	55	55	Complete
Aptimizer V4	Enterprise Collaboration	13	13	Complete
Aptimizer V5	BIM & Smart Construction	13	13	Complete
Aptimizer X	AI Civil Engineering Operating System	55	55	Complete
Aptimizer X+	Future Vision	19	14	In progress

Built and running

Partially built

Planned

Build Status: Where Aptimizer Stands Today

Aptimizer is built and running through **V5** and **Aptimizer X**, with active generative design, autonomous engineering, and smart city infrastructure in **X+**. V1, V2.5, V3, V4, V5, and X are functionally complete. X+ autonomous systems and generative design are operational. Long-range X+ XR/robotics and supplier negotiation capabilities remain planned.

The headline figure: **417 of 424 catalogued features are built**, with a further 1 partially built. Everything still outstanding sits in external live GIS integrations and the long-range XR/robotics and supplier negotiation roadmap.

What was added since the previous feature list

The previous list catalogued 253 features across seven versions. This revision adds 171, with full completion of **V2.5**, **V4**, **V5**, and **Aptimizer X**, and advanced **X+** autonomous and generative capabilities. The largest additions are:

- **Presentation-grade 2D architectural floor plan engine.** Procedural furnished floor plate generation with realistic furniture blocks, structural wall poché (#0F172A), room dimension badges, and multi-format PDF/PNG/ZIP exports.
- **Google Nano Banana AI 3D architectural render engine.** Real-time engineering prompt synthesis from live building geometry, generating photorealistic 3D isometric cutaways, exterior facade views, and luxury interior suites.
- **Site layout engine with genetic refinement.** Multi-objective genetic algorithm (GA) off-grid search, setback envelopes, road reservation, and verified NBC Part 3 spacing clearance.
- **Enterprise collaboration & workflow gating (V4).** 5-stage approval gates with cryptographic audit certificates, multidisciplinary task assignment, threaded reviews, and event-driven automation rules.
- **BIM & Smart Construction Digital Twin (V5).** Native AutoCAD R2018 DWG export, georeferenced IFC4 model export with spatial hierarchy (Revit Link IFC), layered DXF export, IoT sensor streaming with Arrhenius concrete maturity & stripping advisory, 4D BIM progress tracking with EVM (SPI/CPI), and BMS predictive maintenance.
- **Autonomous AI Engineering & Smart City (X+).** Conversational design mutations, 5-agent multi-disciplinary review with consensus certificates, autonomous compliance audit, parametric BOQ takeoff, city-scale zoning, traffic simulation with fire-tender turning clearance, macro utility networks, and urban digital twin microclimate analysis.
- **AI Civil Engineering Operating System (X).** Knowledge graph, memory timeline, decision log audit, what-if decision sandbox, CPWD benchmarks, live material pricing, forecasting, supplier intelligence, JIT calendar, inventory planning, tender documents, government approval dossier, urban growth, multi-hazard risk, green building scorecard, ESG, 30-yr LCC, and executive KPIs.
- **Generative design studio (X+).** Generative facade archetype synthesis with solar metrics and Nano Banana prompts, central park landscape zoning, and automated parking bay layouts.
- **Programme, scheduling, and cash flow.** Activity generation from quantities, full CPM passes, work calendars, prop removal safety rules, a target date solver, cash flow, NPV, and IRR.
- **Data reliability and code intelligence.** Nine per-module data checks, freshness tracking, cross-module consistency, a 74-entry clause registry, and semantic IS/NBC clause verification.

What is not built

- **XR & Robotics and Supplier Negotiation in X+.** AR/VR site walkthroughs, drone survey integration, robotics-assisted construction planning, and AI supplier negotiation.
- **Live External GIS Integrations in V2.** Google Maps live API integration and full satellite imagery basemap tile streaming.

Stated plainly: Aptimizer is a complete enterprise product through Aptimizer X, with full V2.5 precision engineering, complete V5 BIM & Digital Twin smart construction, fully operational Aptimizer X AI Civil Engineering OS, operational X+ autonomous engineering, smart city infrastructure, and generative design.

How the Platform Is Used: Workflow Order

The feature index that follows is ordered by version, because its job is to show what was built when and what comes next. The platform itself is not used in that order. A project moves through five stages, and each stage draws on features from several versions at once. This page maps one onto the other.

Stage	Workspace Module	Draws On	Built
1. Site	Plot & Setbacks	V1, V2.5	Yes
	GIS & Smart City Infrastructure	V2, X+	Yes
2. Design	Autonomous Studio & Township	X, X+	Yes
	Apartment Planning	V1, V2.5	Yes
	Parking Layouts	V1, X+	Yes
	3D Visualisation	V2.5	Yes
	Floor Plans & 3D Deliverables	V1, V2.5, X+	Yes
	BIM, DWG & Digital Twin	V5	Yes
3. Engineering	Calculations & Multi-Agent Teams	V1, X+	Yes
	IS/NBC Compliance Audits	V2.5, X+	Yes
4. Cost & Programme	BOQ Takeoff & Quantities	V1, V2.5, X+	Yes
	Cost Estimation	V1	Yes
	Programme & 4D Progress	V2.5, V5	Yes
	Feasibility & ROI	V2.5	Yes
5. Deliver	Compliance	V1, V2.5	Yes
	Data Reliability	V2.5	Yes
	Reports	V1	Yes
	Versions & Team	V1, V4	Yes

Cutting across every stage

Two capabilities do not sit at a single point in the workflow, which is why they do not appear as a stage of their own.

- **AI assistance (V3).** APT, the optimisers and the AI advisors are reachable from every module, and each draws on the values that module has already computed. The assistant is context-aware by stage rather than being a separate destination.
- **Data reliability (V2.5).** The checks run continuously against every module, but the report is placed in Deliver on purpose. The question it answers is whether the drawing set can be issued, and that is worth asking immediately before the report is generated, not while the design is still moving.

Reading order note: a stage is only as trustworthy as the stage before it. Setbacks decide the buildable envelope, the envelope decides tower placement, placement decides areas, areas decide quantities, quantities decide cost and programme. An error at stage 1 propagates through all five, which is the reason the data reliability checks exist at all.

Aptimizer V1: Core Civil Engineering Planning Platform

Objective: Develop a platform for apartment planning, quantity estimation, cost estimation, and compliance calculations.

126 features | ■ 126 built | ■ 0 partial | ■ 0 planned

User & Project Management

- User Registration & Login
- User Profiles
- Project Dashboard
- Version Comparison
- Project Activity History
- Admin Console
- Role-Based Access Control
- Project Creation
- Project Versioning
- Project Sharing
- Public Compliance Link
- Autosave & Recompute

Plot Management

- Manual Plot Entry
- Plot Dimensions
- Plot Boundary Editing
- Road Access Definition
- Coordinate (Lat/Lng) Entry
- Polygon Plot Creation
- Plot Orientation
- Plot Area Calculation
- Road-Edge Classification
- Polygon Repair & Validation

Apartment Planning

- Apartment Type Selection
- Floor Planning
- Corridor Planning
- Lift Planning
- Common Area Planning
- Procedural Room Layout Generator
- Layout Validation & Retry
- Tower Planning
- Room Planning
- Staircase Planning
- Balcony Planning
- Unit Mix Definition
- Whole-Tower Floor Plate Generation

Civil Engineering Calculations

- Plot Area
- Built-up Area
- Ground Coverage
- FSI Calculation
- Density Calculation
- FAR Derivation Breakdown
- Carpet Area
- Super Built-up Area
- FAR Calculation
- Open Space Calculation
- Area Derivation Audit Trail

Parking Planning

- Basement Parking
- Visitor Parking
- Accessible Parking
- Parking Efficiency Analysis
- Carpet-Band State-Aware Norms
- Shared Site Parking Pool
- Ground Parking
- EV Parking
- Parking Capacity Calculator
- Ramp Planning
- Per-Tower Demand & Supply
- ECS Conversion

Utility Planning

- Water Demand
- Overhead Tank
- WTP Planning
- Electrical Room
- Underground Tank
- STP Planning
- Rainwater Harvesting
- Pump Room

Quantity Estimation

- Concrete
- Steel
- Sand
- Tiles
- Doors
- Plumbing
- Waterproofing
- Cement
- Bricks/Blocks
- Aggregate
- Paint
- Windows
- Electrical
- Finishing Materials

BOQ

- Automatic Bill of Quantities
- Labour Summary
- BOQ PDF Export
- Structure-Derived Quantity Take-off
- Mix Proportion Derivation
- Material Summary
- Equipment Summary
- BOQ Excel Export
- Column & Beam Section Sizing
- Take-off Sanity Checks

Cost Estimation

- Material Cost
- Equipment Cost
- Cost per Flat
- Budget Summary
- Preliminaries
- Contingency
- GST
- Labour Cost
- Construction Cost
- Cost per sq m
- Wastage Allowance
- Overhead & Profit
- Escalation

Compliance Engine

- FAR Validation
- Stair Width
- Lift Compliance
- Accessibility Checks
- Height vs Road Width Rule
- Clause-Referenced Results
- FSI Validation
- Corridor Width
- Fire Exit Checks
- Setback Validation
- Open Space vs Height Rule
- Compliance Margin Warnings

Reports

- BOQ Report
- Quantity Report
- Compliance Report
- PDF Export
- Excel Workbook Export
- Floor Plan Image & ZIP Export
- Cost Report
- Parking Report
- Executive Summary
- Per-Tower Report Sections
- Tower Floor Plans PDF Report

Aptimizer V2: GIS & Site Intelligence

Objective: Integrate maps, satellite imagery, and geospatial analysis.

27 features | ■ 25 built | ■ 1 partial | ■ 1 planned

GIS & Site Intelligence

- | | |
|-------------------------------|-------------------------------|
| ■ OpenStreetMap Support | ■ GIS Coordinate Input |
| ■ Polygon Drawing | ■ Plot Boundary Detection |
| ■ Existing Building Detection | ■ Road Detection |
| ■ Road Width Detection | ■ Green Area Detection |
| ■ Water Body Detection | ■ Terrain Analysis |
| ■ Elevation Analysis | ■ Slope Analysis |
| ■ Flood Risk Analysis | ■ Wind Analysis |
| ■ Sun Path Analysis | ■ Solar Position & Sun Events |
| ■ Annual Insolation | ■ Solar Potential |
| ■ Transit Proximity Analysis | ■ Accessibility Analysis |
| ■ Site Suitability Score | ■ Infrastructure Analysis |
| ■ Buildability Analysis | ■ AI Site Analysis |
| ■ Overpass Feature Caching | ■ Google Maps Integration |
| ■ Satellite Imagery Basemap | |

Aptimizer V2.5: Precision, Programme & Assurance

Objective: Make every number defensible: real development controls, a real site layout engine, a real programme, and a stated confidence in the data behind them.

116 features | ■ 116 built | ■ 0 partial | ■ 0 planned

Development Controls & Setbacks

- Per-Edge Inward Setback Application
- Height-Driven Open Space Table
- Recommended Controls Engine
- Development Control Overrides
- Setback Minimums from Plot Area
- Height vs Road Width Limit
- Setback Verification & Warnings
- Yield Recommendation

Site Layout Engine

- Buildable Envelope Generation
- Local Tangent-Plane Projection
- Perimeter Ring Roads
- Straight-Line Corridor Geometry
- Amenity Block Placement
- Greedy Grid Tower Packing
- Inter-Tower Spacing Enforcement
- Boundary Containment Guarantee
- Layout Recommendation Endpoint
- Genetic Refinement of Placement
- Polygon Repair & Cleaning
- Road Network Reservation
- Internal Spine & Driveways
- Largest Inscribed Block Detection
- Green & Open Space Reservation
- Footprint & Rotation Sweep
- Tower Count & Height Caps
- Shared Fitness Scoring
- Engine Versioning & Signatures

Vastu & Generative Architecture

- Vastu Sector Mapping
- Unit Programme Generation
- MEP Position Audit
- Vastu-Aligned Floor Plates
- Plan Realism Check
- Sector-Anchored Unit Packing
- Vastu Audit per Unit
- Sector Report
- Architectural Template Generation
- Room Aspect & Dimension Checks

Programme & Scheduling

- Activity Generation from Quantities
- Critical Path Identification
- Float Verification
- Duration from Output Rates
- Wage Premium Modelling
- Schedule Safety Audit
- Task Overrides & Edits
- Cash Flow Curve
- Line of Balance
- Gantt Visualisation
- CPM Forward & Backward Pass
- Total & Free Float
- Work Calendar & Holidays
- Crew Efficiency Modelling
- Prop Removal Safety Rules
- Scheduled Date Audit
- Target Date Solver
- Resource Histogram
- Milestones

Development Finance

- Saleable Area Derivation
- S-Curve Cost Spread
- NPV
- Break-even Point
- Finance Configuration
- Revenue Model
- Monthly Cash Flow
- IRR
- Preliminaries per Month

Data Reliability & Assurance

- Plot Data Check
- Tower Data Check
- GIS Data Check
- Engineering Check
- Programme Check
- Cross-Module Consistency Checks
- Module Warning Rollup
- Data Health Report Export
- Development Controls Check
- Site Layout Check
- Parking Check
- Finance Check
- Data Freshness Tracking
- Compliance Margin Reporting
- Overall Trust Score

IS & NBC Code Intelligence

- Central Code Constant Registry
- Clause Text Corpus Ingestion
- Semantic Clause Retrieval (RAG)
- Code Question Answering
- Citation Verification Against Registry
- 74-Entry Clause Registry
- Clause Chunking & Indexing
- Code Search Endpoint
- Citation Extraction
- Cited Extracts in Answers

Engineering Modules (IS/NBC)

- Structural Load Estimator (IS 875)
- Seismic Zone & Base Shear (IS 1893)
- Concrete Mix Design (IS 10262)
- Storm Water & RWH (IS 3764)
- Fire Safety Compliance (NBC Part 4)
- Green Building Preliminary Rating
- Embodied Carbon
- Wind Force Coefficients (IS 875-3)
- Foundation Advisor (IS 6403/1904)
- Water Infrastructure (IS 1172)
- Parking Compliance (NBC/SP:21)
- Accessibility Compliance (NBC Part 3)
- Column Grid Optimizer
- Plantation Plan

3D & Visualisation

- 3D Site Model
- Road & Plot Rendering
- Central Park & Landscape
- Site Diagrams
- Live Metrics Strip
- Presentation-Grade Furnished 2D Blueprint Engine
- Tower Massing
- Amenity & Clubhouse Rendering
- Floor Plate Viewer
- Blueprint View
- Command Palette
- Structural Wall Poché & Dimension Badges

Aptimizer V3: AI Engineering Platform

Objective: Introduce AI-driven planning and decision support.

55 features | ■ 55 built | ■ 0 partial | ■ 0 planned

AI Planner

- Automatic Layout Generation
- Apartment Mix Optimisation
- Open Space Optimisation
- Smart Tower Placement
- Floor Optimisation

AI Civil Engineer

- Civil Design Suggestions
- FSI Optimisation
- Parking Optimisation
- FAR Optimisation
- Utility Optimisation

AI Structural Advisor

- Column Grid Suggestions
- Foundation Recommendations
- Beam Layout Suggestions
- Structural Risk Warnings

AI Quantity Surveyor

- Quantity Optimisation
- Steel Cutting Optimisation
- BOQ Optimisation
- Waste Reduction
- Tile Waste Optimisation

AI Cost Engineer

- Cost Prediction
- Cost Comparison
- Grade Sweep Analysis
- Budget Optimisation
- Cost Saving Suggestions

AI Sustainability

- Green Building Score
- Solar Potential
- Tree Plantation Suggestions
- Carbon Footprint
- Water Efficiency

AI Business Consultant

- ROI Analysis
- Project Feasibility
- Profit Forecast
- Break-even Analysis

AI Recommendation Engine

- Design Improvements
- Cost Optimisation
- Engineering Suggestions
- Material Recommendations

APT: In-App AI Assistant

- Project Q&A;
- Report Explanation
- Streaming Responses
- Project State Context Builder
- Query Classification
- Chat Threads & History
- Provider Fallback & Retry

Scenario Comparison

- Compare Layouts
- Compare Parking
- Compare ROI

- Civil Engineering Guidance
- Scenario Comparison
- Module-Aware Opening Questions
- Context & Analysis Caching
- Capability Questions
- Citation Verification

- Compare Costs
- Compare Sustainability

Aptimizer V4: Enterprise Collaboration

Objective: Enable teamwork and enterprise workflows.

13 features | ■ 13 built | ■ 0 partial | ■ 0 planned

Collaboration & Workflow

- | | |
|---|---|
| ■ Multi-user Collaboration | ■ Project Sharing |
| ■ Team Roles | ■ Version Control |
| ■ Activity Logs | ■ Task Assignment |
| ■ Approval Workflow | ■ Comments & Reviews |
| ■ Workflow Automation | |

Engineering Decision Support

- | | |
|--|---|
| ■ Construction Timeline | ■ Labour Planning |
| ■ Equipment Planning | ■ Utility Network Planning |

Aptimizer V5: BIM & Smart Construction

Objective: Prepare for BIM, Digital Twin, and construction monitoring.

13 features | ■ 13 built | ■ 0 partial | ■ 0 planned

BIM Readiness

- | | |
|--|--|
| ■ IFC Export | ■ DWG Export |
| ■ DXF Export | ■ Revit Compatibility |
| ■ AutoCAD Compatibility | |

Digital Twin Preparation

- | | |
|---|---|
| ■ IoT Readiness | ■ Smart Sensor Integration |
| ■ Progress Tracking | ■ Quality Monitoring |
| ■ Safety Monitoring | ■ Delay Prediction |
| ■ Predictive Maintenance | ■ Facility Management |

Aptimizer X: AI Civil Engineering Operating System

Objective: A unified AI-powered platform for the entire project lifecycle.

55 features | ■ 55 built | ■ 0 partial | ■ 0 planned

AI Operating System

- AI Engineering Copilot
- Engineering Knowledge Graph
- Engineering Decision Log
- AI Engineering Consultant
- Engineering Memory
- Explainable AI

Intelligent Planning

- One-Click Project Generation
- AI Township Planner
- Apartment Capacity Prediction
- Smart Building Recommendations
- AI Layout Generator
- Mixed-Use Development Planning
- Land Suitability Analysis

GIS & Urban Intelligence

- City Growth Prediction
- Infrastructure Growth Prediction
- Disaster Risk Analysis
- Flood Analysis
- Noise & Pollution Analysis
- Land Value Prediction
- Climate Analysis
- Earthquake Analysis
- Wind Analysis

Smart Procurement

- Live Material Prices
- Supplier Intelligence
- Inventory Planning
- Material Price Forecasting
- Procurement Calendar

Financial Intelligence

- Cash Flow Planning
- Investment Analysis
- Profitability Dashboard
- Lifecycle Cost Analysis
- ROI Forecasting

Sustainability

- Green Building Planner
- Solar Energy Planning
- ESG Reporting
- Carbon Emission Calculator
- Water Conservation Analysis

Construction Intelligence

- Construction Timeline Optimisation
- Labour Optimisation
- Construction Delay Prediction
- Resource Allocation
- Equipment Scheduling
- Risk Prediction

Reports & Decision Support

- AI Report Writer
- Interactive Analytics
- Scenario Simulation
- Executive Dashboards
- Decision Sandbox
- Engineering Benchmarking

Marketplace & Ecosystem

- Engineering Marketplace
- API Marketplace
- Government Approval Assistant
- Plugin Marketplace
- Educational Mode
- Tender Document Generator

Aptimizer X+: Future Vision

Objective: A future roadmap beyond Aptimizer X.

19 features | ■ 14 built | ■ 0 partial | ■ 5 planned

Autonomous AI Engineering

- Conversational Project Design
- Autonomous Compliance Checking
- AI Negotiation with Suppliers
- Multi-Agent Engineering Teams
- Autonomous BOQ & Costing

Generative Design

- AI-Generated 2D Floor Plans
- Facade Design Suggestions
- Parking Layout Generation
- AI-Generated 3D Building Concepts
- Landscape Design

Smart City Platform

- City-Scale Planning
- Utility Network Optimisation
- Infrastructure Demand Forecasting
- Traffic Simulation
- Urban Digital Twin

XR & Robotics

- AR Site Visualisation
- Drone Survey Integration
- VR Project Walkthroughs
- Robotics-Assisted Construction Planning